

can be unstable in noisy data

allow diffusive noise

Motivation and Intuition

- Dynamic OT gives sharp displacement but **can be unstable in noisy data**.
- What if we **allow diffusive noise** in the paths?
- Schrödinger bridge (SB): “Most likely stochastic process (close to Brownian motion) that matches observed marginals.”
- ML connections:
 - Diffusion Schrödinger Bridge (De Bortoli et al. 2021, etc.)
 - Score-based diffusion models ~ SBs with learned guidance.

Dynamic Formulation on Path Space